METRIC

DOD-B-24506/1C(SH)
20 March 1978
SUPERSEDING
MIL-B-24506/1B(SH)
1 June 1976

MILITARY SPECIFICATION SHEET

BATTERY, STORAGE, SILVER-ZINC ALKALINE TYPE

(FOR TRIESTE) PROPULSION (1000 AMPERE-HOUR) (METRIC)

This specification sheet is approved for use by the Naval Sea Systems Command and is available for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the battery described herein shall consist of this document and the latest issue of DOD-B-24506.

REQUIREMENTS:

Electrical characteristics: See table I.

TABLE I - Electrical characteristics.

Discharge rate ampere (A)	Time hours (h)	Capacity ampere-hour (Ah)	Battery minimum voltage (V)	Minimum average 1/ cell voltage (V)	Minimum (cut-off) cell voltage (V)
405	2	810	100	1.44	1.20
225	4	900	106	1.46	1.25
125	8	1000	109	1.48	1.30
50	20	1000	110	1.50	1.34

1/ The "average voltage" is defined as the average of voltage recordings taken at equal intervals, starting with the first recording (taken between 1/2 minute and 1 minute after applying a constant rate) and including the last recording. The interval of recordings shall be not greater than h/5, where h is the time in hours of the rated capacity of the test. The time to the last recording shall be not greater than the rated time.

Life: 24 months or 80,000 Ah, whichever occurs first.
Cell dimensions - millimeters (mm):
Maximum length: 137 mm.
Maximum width: 137 mm.
Maximum height (including electrolyte entrainment eliminators: 513 mm.
Cell mass: 19 kilograms (kg) (maximum); 18.5 kg (average).
Cell jar: See figure 1.
Cell cover: See figure 2.
Battery assembly: See figures 3, 4, 5, and 6.
© Total cells offered for delivery: 94.
(a) First article cells: 3.
(b) Battery cells: 80.
(c) Spare cells: 10.
(d) Sample cells for quality conformance: 3.
K factor for quality conformance (capacity): 3.1.
© Hydrogen evolution: maximum 20 ml/hour.
Charge efficiency: For determination of charge efficiency curve.
(a) All discharges at the 8-hour rate.
(b) Charges at the 50, 100, 150, 200, 300, and 400-ampere rates.

Preparing activity:
Navy - SH
(Project 6140-N474-1)

© denotes changes

FSC 6140

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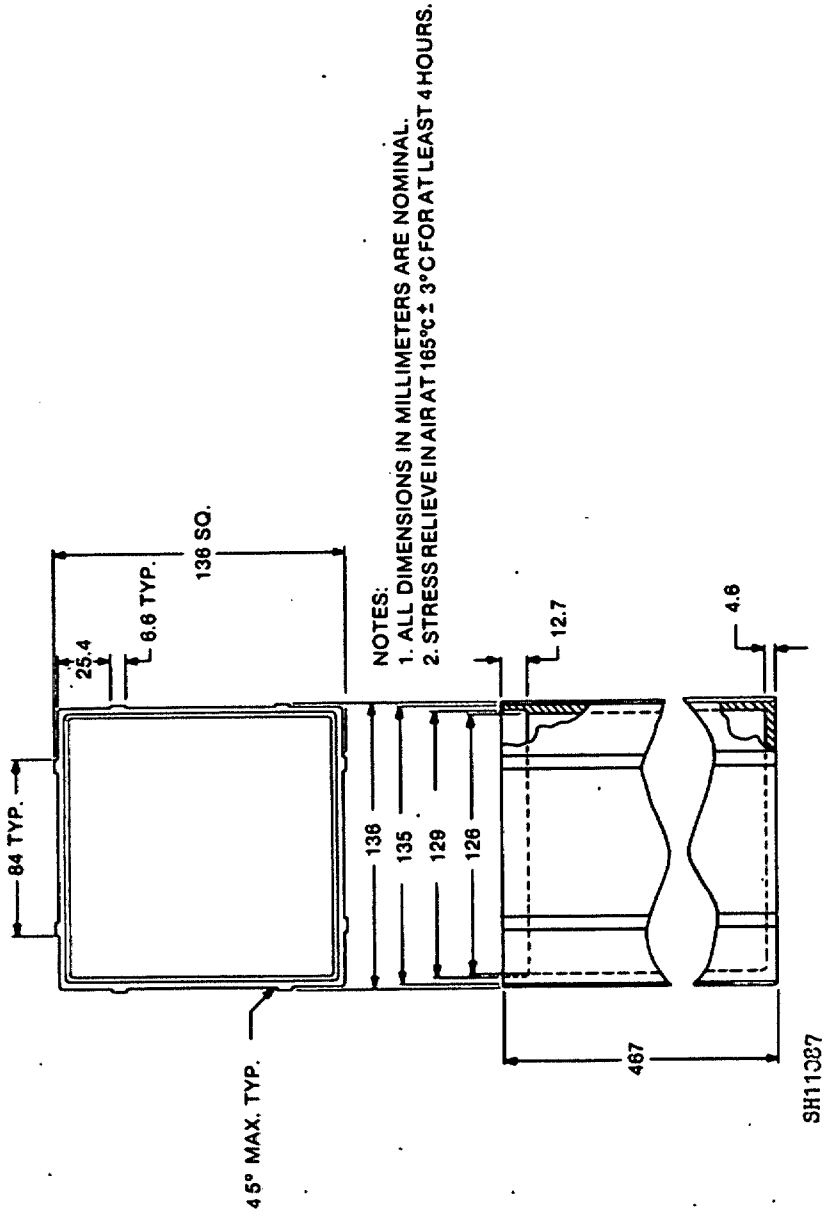
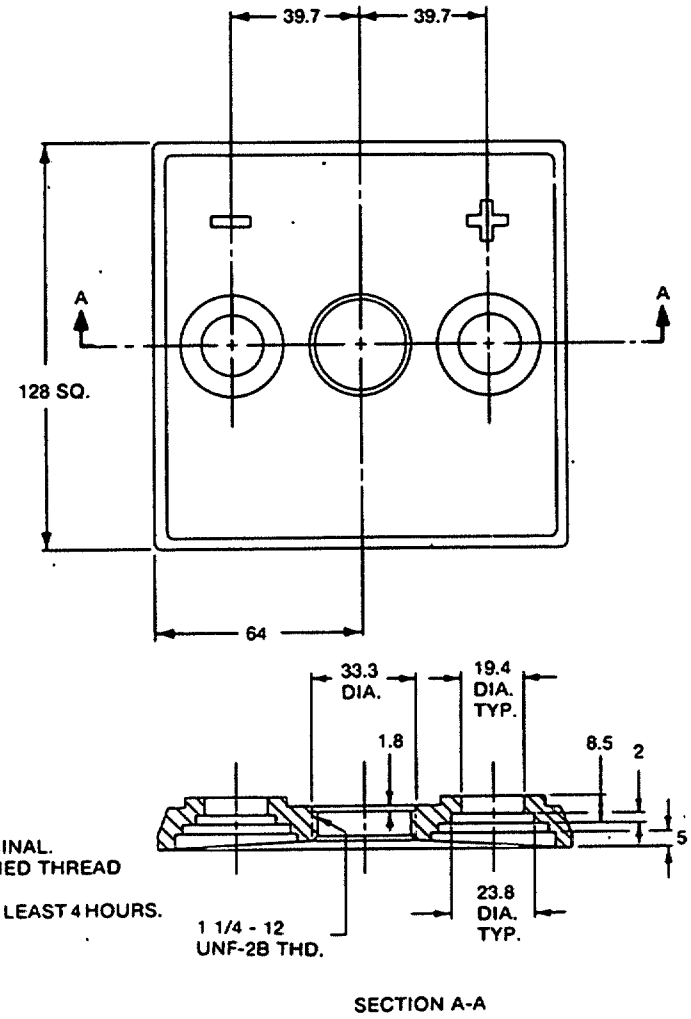


FIGURE 1. Cell cover.

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NOTES:
1. ALL DIMENSIONS IN MILLIMETERS ARE NOMINAL. THREADED HARDWARE DESIGNATION IS UNIFIED THREAD STANDARD.
2. STRESS RELIEVE IN AIR AT 165°C ± 3°C FOR AT LEAST 4 HOURS.

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FIGURE 2. Cell cover.

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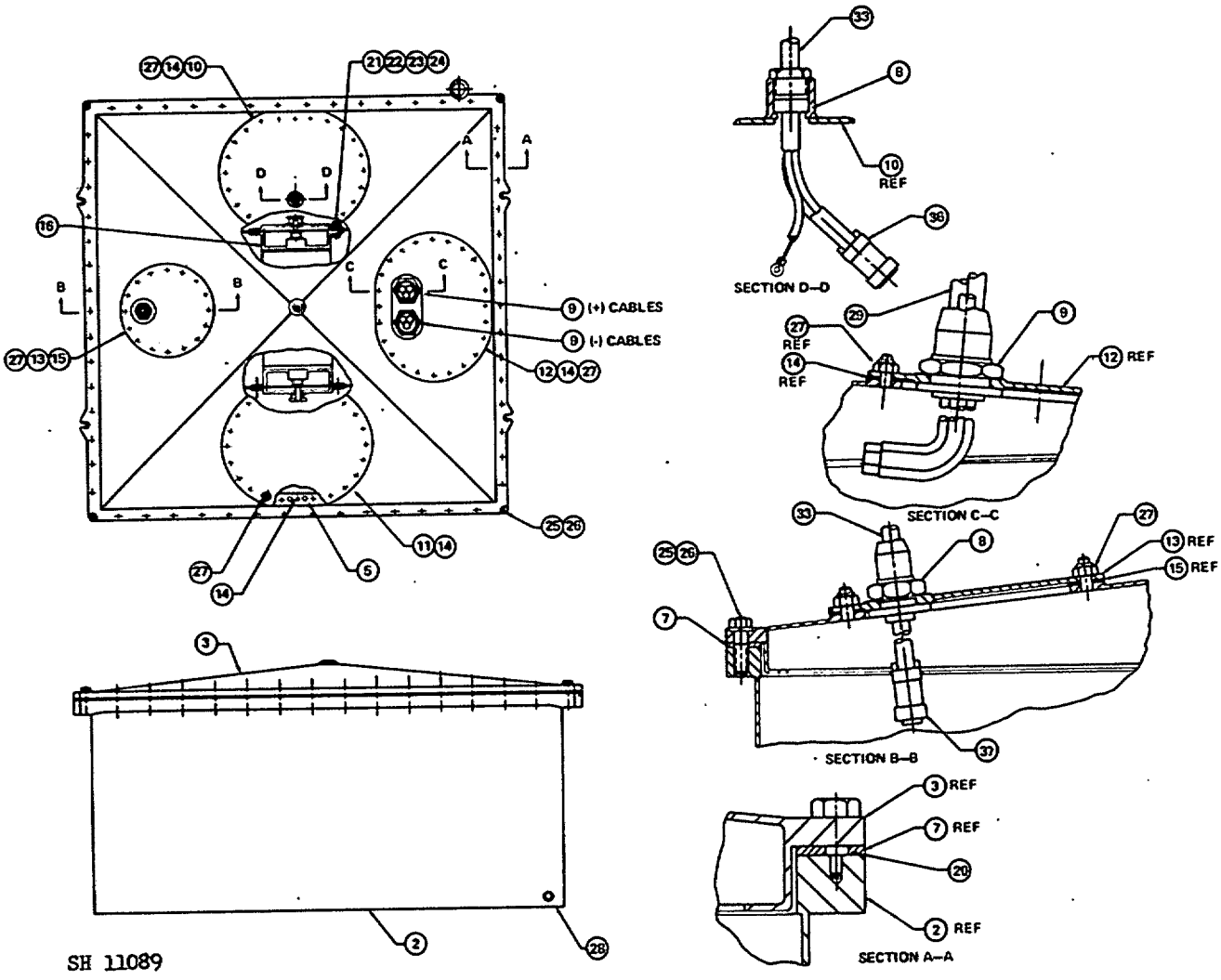


FIGURE 3. Battery assembly (battery box and cover).

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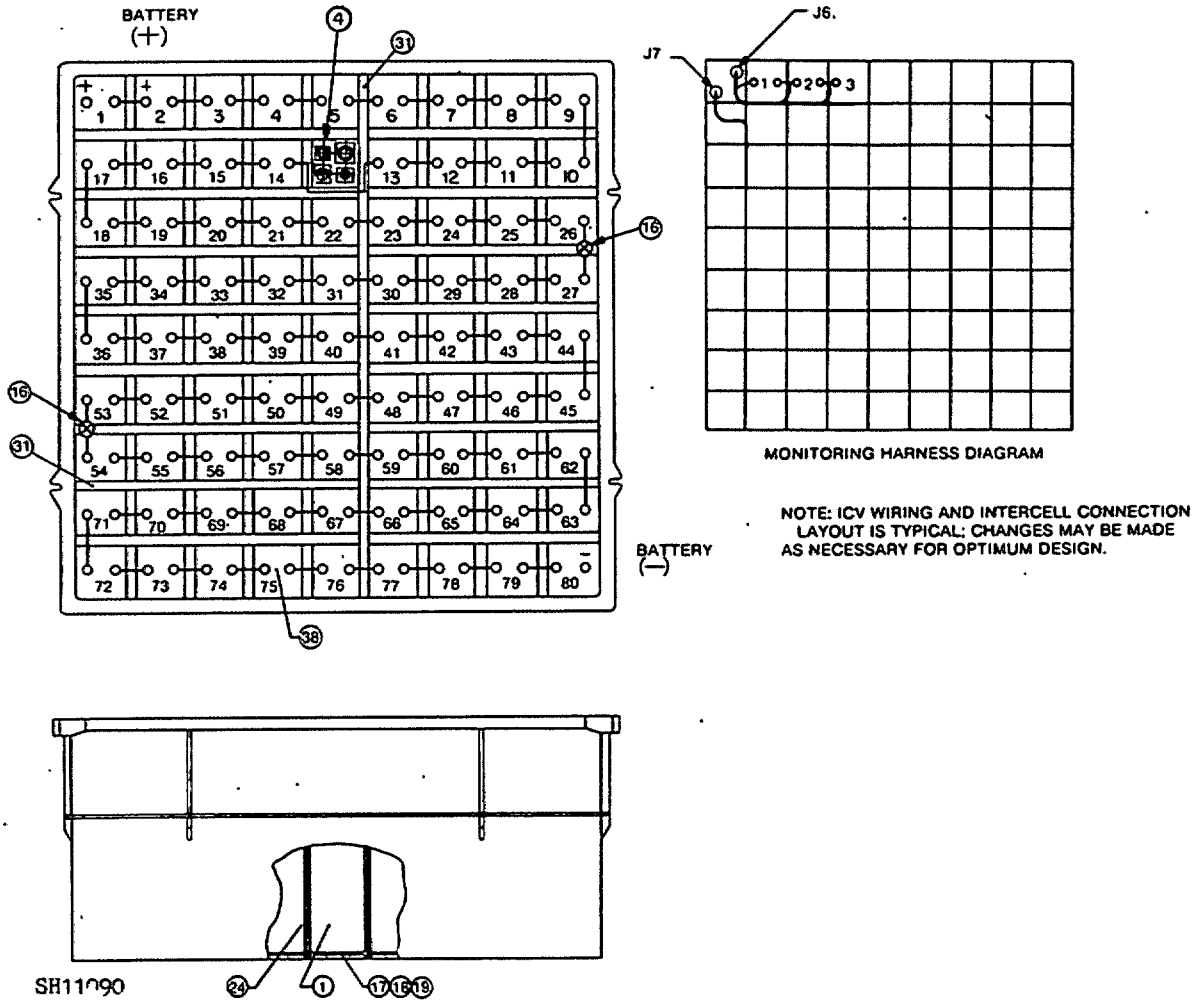


FIGURE 4. Battery assembly (typical layout).

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Item no.	Qty reqd	Nomenclature or description
1	80	Cell
2	1	Case, battery (120V)
3	1	Cover, battery (120V)
4	1	Scanner switch assembly
5	88	Spacer, gasket
6	320	Spacer, cell
7	1	Gasket
8	2	Tube, stuffing
9	2	Tube, stuffing
10	1	Cover, manhole
11	1	Cover, manhole
12	1	Cover, manhole
13	1	Cover, manhole
14	3	Gasket manhole
15	1	Gasket, manhole
16	2	Disconnect, (120V)
17	16	Pad, support
18	4	Pad, support
19	1	Pad, support
20	60	Spacer, gasket
21	4	Nut, hex 1/2-13UNC (CRES)
22	8	Flat, washer, 13 mm (CRES)
23	4	Lockwasher, 13 mm (CRES)
24	4	Bolt, hex HD (CRES 303) 1/2-13UNC x 32 mm LG.
25	60	Bolt, hex HD (NI-CU) 5/8-11UNC x 50 mm LG.
26	60	Lockwasher, 16 mm (CRES)
27	80	Locknut, 1/2-13UNC
28	1	Plug, drain 3/4 NPT
29	---	Cable
30	50	Strap, tie-down, self-locking
31	AR	Tape, elec (vinyl) 19 mm wide
32	AR	Wire, elec -C16AWG, blk
33	1	Cable, MWF-19 or equal
34	1	Connector, elec, MS3106E-28-15S
35	1	Connector, elec, MS3106E-36-10S
36	1	Connector, elec, MS3106E-22-14S
37	1	Connector, elec, MS3106E-20-29S
38	80	Disc, ID (1-80)

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NOTE: DIMENSIONS ARE IN MILLIMETERS. THREADED HARDWARE DESIGNATION IS UNIFIED THREAD STANDARD.

FIGURE 5. Battery assembly (key to figures 3 and 4).

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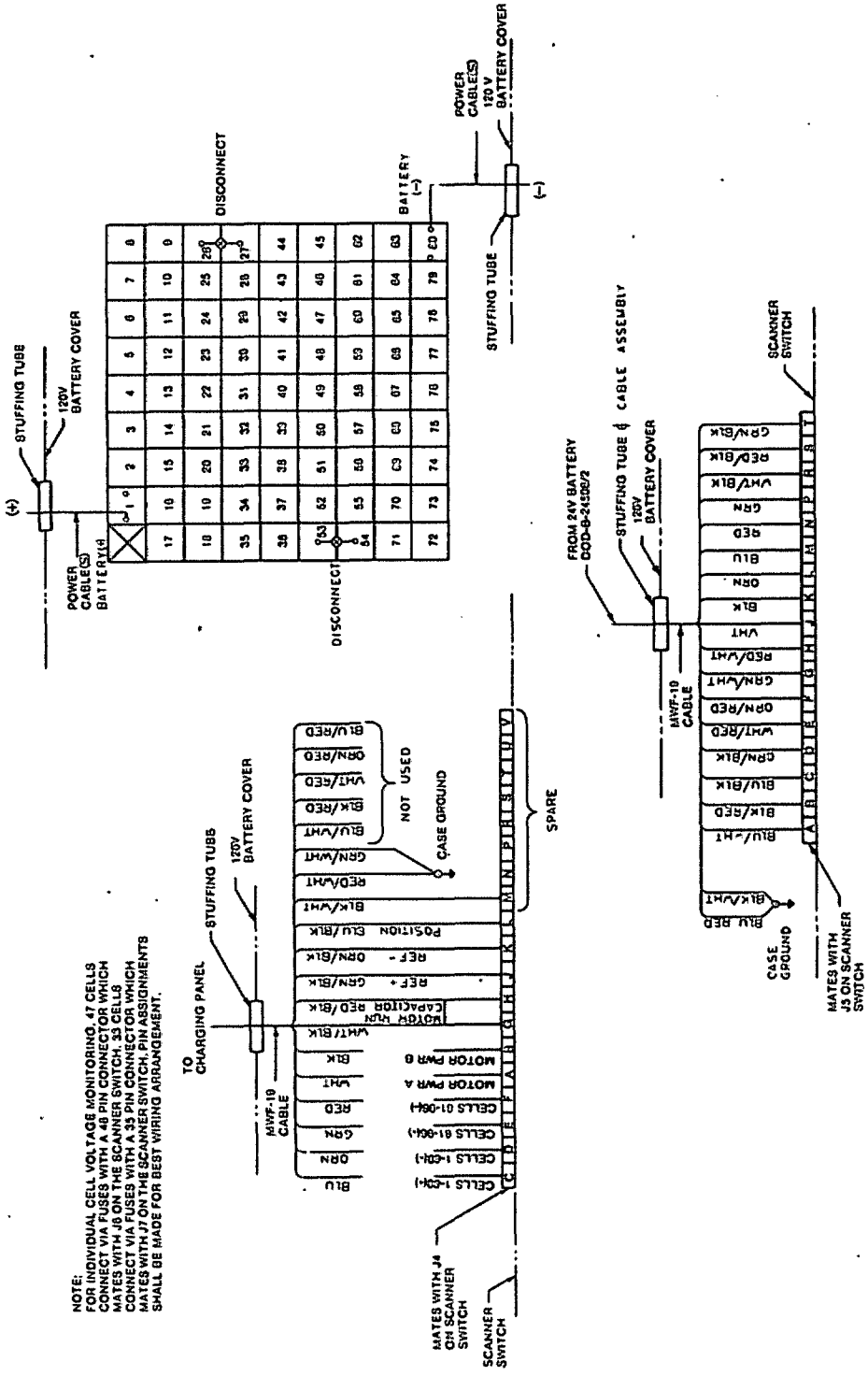


FIGURE 6. Battery assembly (wiring diagram).

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No. req'd cable or wire	From	To	Length ± 75 mm
(1) MWF-19	Stuffing tube	Connector, MS3106E-22-14S	As required to allow cover to lift 1 meter
(1) MWF-19	Stuffing tube	Connector, MS3106E-20-29S	As required to allow cover to lift 1 meter
(1) MWF-19	Stuffing tube	Charging panel	Not less than 15 meters
(1) MWF-19	Stuffing tube	24 volt battery	3 meters
Power cable	Disconnect	Cells 28, 29, 53 and 54	As required to allow cover to lift 1 meter
Power cable	Stuffing tube	Cell 80(-)	As required to allow cover to lift 1 meter
Power cable	Stuffing tube	Cell 1(+)	As required to allow cover to lift 1 meter
Power cable	Stuffing tube	120 volt (+) breaker	3 meters
Power cable	Stuffing tube	120 volt (-) breaker	3 meters
Wire blk 16 AWG	Cells 1-47	Connector, MS3106-36-10S	As required
Wire blk 16 AWG	Cells 48-80	Connector, MS3106-28-15S	As required

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FIGURE 6. Battery assembly wiring diagram (con.).

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